



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing
SECI 1013 – Discrete Structure

Prepared for Dr. Shukor Talib

ASSIGNMENT 3

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Question 1

- a) Ali has 5 certificates to be stored in 3 different folders. Show that at least 2 certificates share the same folder.

pigeonholes = 3 folders

$$k = \left\lceil \frac{n}{m} \right\rceil$$

pigeon = 5 certificates

$$k = \left\lceil \frac{5}{3} \right\rceil$$

$$k = 2$$

folders 1 : 1 certificates

folders 2 : 1 certificates

folders 3 : 1 certificates

Since there's still 2 certificates left, therefore there is a folder that contain at least 2 certificates.

- b) In a group of 20 people, each person has one pet that is a cat, a dog or a goat. What is the minimum numbers of people you need to be sure two of them have the same type of pet.

pigeonholes = cat, dog, goat (~~four~~ types of pet)

pigeon : people

The maximum number of people who can all have different pet types is 3, which is one dog, one cat and one goat.

to at least guarantee two people have the same type of pet is to add one more person :-

$$\text{minimum number} = (\text{number of type of pet}) + 1$$

$$= 3 + 1$$

$$= 4$$

∴ therefore minimum number is 4

Question 2

(a) A 4 digit code is formed using digits 0-9, repetition allowed.

Find the probability that it contains exactly two distinct digits.

Total possible codes =

$$10^4 = 10000$$

Codes with exactly two distinct digits.

1. Choose 2 different digits from 10:

$$\binom{10}{2} = 45$$

2. Arrange them in 4 positions but not all same digit:

$$2^4 - 2 = 16 - 2 = 14$$

Total favourable outcomes:

$$45 \times 14 = 630$$

probability

$$p = \frac{630}{10000} = 0.063$$

(b) Given the probability mass function:

x	0	1	2	3
$p(x)$	0.1	0.3	0.4	0.2

(i) Find $P(X \geq 2)$

$$P(X \geq 2) = P(2) + P(3)$$

$$= 0.4 + 0.2 = 0.6$$

(ii) Find $P(3 > X \geq 0)$

Values of X : 0, 1, 2

$$p = P(0) + P(1) + P(2)$$

$$= 0.1 + 0.3 + 0.4 = 0.8$$

Question 3

A shipment contains defective items with a probability of 0.05.

- A defective item is returned with probability 0.9

- A non-defective item is returned with probability 0.1

9) Find the probability that a randomly selected item is returned.

D - defective

N - Non-defective

R - returned

$$P(D) = 0.05$$

$$P(N) = 1 - 0.05 = 0.95$$

$$P(R|D) = 0.9$$

$$P(R|N) = 0.1$$

$$P(R) = P(R \cap D) + P(R \cap N)$$

$$P(R) = P(R|D)P(D) + P(R|N)P(N)$$

$$P(R) = (0.9)(0.05) + (0.1)(0.95)$$

$$P(R) = 0.14$$

b) Given that an item is returned, find the probability that it is defective.

$$P(D|R) = \frac{P(R|D)P(D)}{P(R)}$$

$$P(D|R) = \frac{(0.9)(0.05)}{0.14}$$

$$P(D|R) = \frac{0.045}{0.14}$$

$$P(D|R) = 0.3214$$

Q4

BB BG GB GG

E: at least one boy

F: boy and girl

G: firstborn is boy

a) $P(E)$?

$$E = \{BB, BG, GB\}$$

$$\begin{aligned} P(E) &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} \\ &= \frac{3}{4} \end{aligned}$$

b) $P(F)$?

$$F = \{BG, GB\}$$

$$\begin{aligned} P(F) &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

c) $P(E \cap F)$?

$$\begin{aligned} P(E) \cap P(F) &= \{BG, GB\} \\ &= \frac{1}{4} + \frac{1}{4} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} d) P(E) \cdot P(F) &= \frac{3}{4} \times \frac{1}{2} \\ &= \frac{3}{8} \end{aligned}$$

$$P(E \cap F) = \frac{1}{2}$$

$$P(E \cap F) \neq P(E) \cdot P(F)$$

$\therefore$ Since $P(E \cap F)$ is not equal to $P(E) \cdot P(F)$,
thus E and F are not independent.

Q5

$$A - 40\% - 2\%$$

$$B - 35\% - 4\%$$

$$C - 25\% - 1\%$$

a) defective probability

$$\begin{aligned} P(D) &= (0.4)(0.02) + (0.35)(0.04) + (0.25)(0.01) \\ &= 0.008 + 0.014 + 0.0025 \\ &= 0.0245 \end{aligned}$$

b)

$$P(B|D) = \frac{P(B \cap D)}{P(D)}$$

$$\begin{aligned} P(B \cap D) &= P(B) \cdot P(D|B) \\ &= 0.35 \times 0.04 \\ &= 0.014 \end{aligned}$$

$$\begin{aligned} P(B|D) &= \frac{0.014}{0.0245} \\ &= 0.571 \end{aligned}$$

Question 6

Let

S = Spam

L = Legitimate

F = Email flagged as spam

Given :

$$P(S) = 0.15, \quad P(L) = 0.85$$

$$P(F|S) = 0.95, \quad P(F|L) = 0.08$$

(a) Find $P(F)$

$$P(F) = P(F|S)P(S) + P(F|L)P(L)$$

$$= (0.95)(0.15) + (0.08)(0.85)$$

$$= 0.1425 + 0.068$$

$$= 0.2105$$

$$(b) P(S|F) = \frac{P(F|S)P(S)}{P(F)}$$

$$= \frac{0.95 \times 0.15}{0.2105}$$

$$= \frac{0.1425}{0.2105}$$

$$\approx 0.677$$